Fiscal Year:	FY 2009	**Task Last Updated:**	FY 08/07/2009
PI Name:	Hall, Eric J Ph.D., D.Sc.		
Project Title:	Individual Genetic Susceptibility		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	06/01/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Correct end date is 6/1/2009 per C. Guidry/JSC (2/2010) NOTE: Project to be extended into 2010, per J. Dardano; details of extension not yet available (7/2009) NOTE: Received NCE to 05/31/2009 per PI (7/2008) NOTE: Received NCE to 6/1/2008 per P. Denkins (6/07) NOTE: end date changed to 5/31/2007 per J. Dardano (5/06)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Brenner, David J (Columbia University) Smilenov, Lubomir (Columbia University)		
Grant/Contract No.:	NAG 9-1519		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The hypothesis upon which this proposal is based is that the human population is not homogeneous in radiosensitivity, but that radiosensitive sub-groups exist which would suffer an increased incidence of detrimental radiation effects, and distort the shape of the dose response relationship. The radiosensitivity of these groups depends on the expression levels of specific proteins. In a previous proposal funded by NASA we used knock-out mice to show unequivocally that haploinsufficiency for the ATM gene resulted in increased sensitivity to gamma-rays for both ocular cataracts (a deterministic effect) and oncogenic transformation in embryo fibroblasts (a stochastic effect). It would be extraordinarily difficult and expensive to conduct similar experiments with the high energy Fe-59 ions of principal interest to NASA, because of the need to breed a large number of animals, all of the appropriate age, to be available on the few occasions each year when Fe-59 ions are available at the Brookhaven Laboratory. It would be even more difficult and expensive to expand the study to include a number of other genes that are suspected of being involved with radiosensitivity, and to investigate the possible interaction of these various genes. We proposed instead to screen cells heterozygous for a number of genes; Atm, BRCA1, BRCA2 and rad9, as well as double heterozygotes for pairs of these genes. Wild type animals as well as heterozygotes and double heterozygotes will be exposed to graded doses of ?-rays at Columbia or Fe-56 heavy ions at Brookhaven National Laboratory, and thymocytes assayed for early apoptotic death. By this means we plan to discover which genes or combinations of genes, are important in conferring radiosensitivity. For genes, or combinations of genes that look promising for this apoptotic endpoint, more extensive studies would be carried out with the more relevant endpoint oncogenic transformation using the embryo fibroblasts. This strategy is based on preliminary studies with gamma-rays and cell lines with a mutated gene (namely Atm) which show that a reduction in radiation-induced apoptosis is mirrored by an increase in oncogenic transformation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The dose limits proposed for astronauts in space flight are different from those on the ground, though the basic philosophy of limiting the life-time risk of stochastic effects to 3% is preserved. There are several reasons for the differences, including the limited number of individuals exposed, the practicality of shielding and the presence of more urgent risks. Nevertheless the purpose of radiation protection remains to prevent deterministic effects of clinical significance and limit stochastic effects to acceptable levels. The levels proposed are based on the assumption that the human population is uniform in its radiosensitivity. Using two endpoints, namely cataracts as an important and relevant deterministic effect and oncogenic transformation in cultured embyro cells as a surrogate for carcinogenesis, the most relevant stochastic effect, we have shown that mice heterozygous for the atm gene are more radiosensitive than wild type animals. 1 to 3% of the US population are AT heterozygotes, and it would not be prudent to subject such individuals to substantial doses of radiation.
Task Progress:	Radiation exposures in Space are relatively difficult to reduce, and impossible to eliminate entirely. At the same time, other risks to humans in the hostile environment in space may be more acute than those of radiation. For this reason, larger annual dose limits have been tolerated for astronauts than recommended by NCRP for radaiation workers on the ground, though career limits of risk have been roughly equalized. There are now two clear instances from human epidemiological studies which suggest that the human population is not homogeneous in its response to radiation, but rather that radiosensitive sub-groups exist. One involves the clustering of meningiomas in certain north African families where children were epilated with x-rays in the course of the treatment of Tinea Capitis. The other involves a study of breast cancer in young people monitored by frequent diagnostic x-rays in the management of scoliosis. In this case, the dose-response was significantly greater for women who reported a family history of breast cancer in first or second degree relatives While in each of these instances there is a clear indication of a radiosensitive sub-group, the gene(s) involved were not identified though there were obvious suspects, ATM in the case of the North African children and BRCA1 in the case of the young people with scoliosis. Against this background we chose to study the effect on the biological response to heavy particles, similar to those found in space of cells heterozygous for three low frequency, high penetrance genes, namely Atm, Brca1 and Rad9. This study was possible because knock-out mice, heterozygous for each of these genes, were available in our laboratory. Furthermore, by cross-breeding these animals we were able to obtain animals heterozygous for pairs of these genes. Fibroblasts from mid-term embryos were harvested from all of these animal types, irradiated with heavy ions at Brookhaven, and the incidence of oncogenic transformat ion assessed.(this is an in vitro surrogate for carcinogenesis.) At the same time, thymocyes were harvested from each animal type, irradiated at Brookhaven, and the incidence of early cell death by apoptosis assayed. The results are interesting. For all three genes, the incidence of radiation induced transformation was greater in heterozygous than in wild type cells, and conversely the incidence of radiation-induced apoptosis was lower in heterozygous cells. The reduced apoptotic levels associated with heterozygous genotypes shows that more cells with unrepaired DNA survive, leading to increased transformation. Mutations and/or polymorphisms are present in the homologues of these genes in the human population, suggesting an explanation for the presence of a radiosensitive sub-population of approximately 5%. The study of the double heterozygotes also raises an interesting point. The cells heterozygous for both the Atm and Rad9 genes were more radiosensitive than cells heterozygous for Atm or rad9 alone, suggesting an additive effect of the two genes. By contrast, cells heterozygous for both Atm and Brca1 were no more radiosensitive than cells heterozygous for Atm alone. A possible explanation for this difference is that the Atm and Brca1 genes operate in the same pathway, whereas Atm and Rad9 do not. There are two important consequences of the existence of a radiosensitive sub-population. First it would distort the shape of the dose-response relationship for end-points such as carcinogenesis. Linearity between dose and effect would no longer be the case if a fraction of the population is radiosensitive. Second, it would be unethical to place a radiosensitive individual in a situation where they may be exposed to a significant dose of radiation. It is entirely possible that other genes will be identified that could add to the size of the radiosensitive sub-population.
Bibliography Type:	Description: (Last Updated: 10/26/2023)

Articles in Peer-reviewed Journals	Hall EJ. "Is there a place for quantitative risk assessment? " Journal of Radiological Protection, 2009 Jun;29(2A):A171-84. PubMed PMID: 19454800 , Jun-2009
Articles in Peer-reviewed Journals	Su F, Smilenov L, Ludwig T, Zhou L, Shu J, Zhou G, Hall EJ. "Heterozygosity for Atm and Brca1 influence the balance between cell transformation and apoptosis." International Journal of Radiation biology (submitted), May 2009. , May-2009
Articles in Peer-reviewed Journals	Zhou G, Smilenov LB, Lieberman HB, Hall EJ. "Radiosensitivity to high energy iron ions is influenced by heterozygosity for Atm, Rad9 and Brca1." Advances in Space Research. (Submitted). June 2009. , Jun-2009
Articles in Peer-reviewed Journals	Kleiman NJ, David J, Elliston CD, Hopkins KM, Smilenov LB, Brenner DJ, Worgul BV, Hall EJ, Lieberman HB. "Mrad9 and Atm haploinsufficiency enhance spontaneous and X-ray-induced cataractogenesis in mice." Radiat Res. 2007 Nov;168(5):567-73. PMID: 17973559 , Nov-2007
Articles in Peer-reviewed Journals	Suit H, Goldberg S, Niemierko A, Ancukiewicz M, Hall E, Goitein M, Wong W, Paganetti H. "Secondary carcinogenesis in patients treated with radiation: a review of data on radiation-induced cancers in human, non-human primate, canine and rodent subjects." Radiat Res. 2007 Jan;167(1):12-42. Erratum in: Radiat Res. 2007 Jun;167(6):748. PubMed PMID: 17214511 , Jan-2007
Articles in Peer-reviewed Journals	Kleiman NJ, Edmondson EF, Weil MM, Fallgren CM, King A, Schmidt C, Hall EJ. "Radiation cataract in heterogeneous stock mice after ?-ray or HZE ion exposure." Life Sci Space Res. 2023 Sep 15. https://doi.org/10.1016/j.lssr.2023.09.004 , Sep-2023